

Palo Alto Firewall Deployment in AWS using Infrastructure as Code (Terraform)



Project Overview

This guide provides a detailed walkthrough for deploying a Palo Alto Network VM series firewall in AWS cloud using Terraform

Required Tools

- AWS Account
- Terraform (v1.5.0+)
- AWS CLI
- SSH Key Pair

Architectural Components

- VPC with multiple subnets
- Palo Alto VM-Series Firewall
- Management and Data Interfaces
- Elastic IP Configuration
- Security Groups



Project Structure

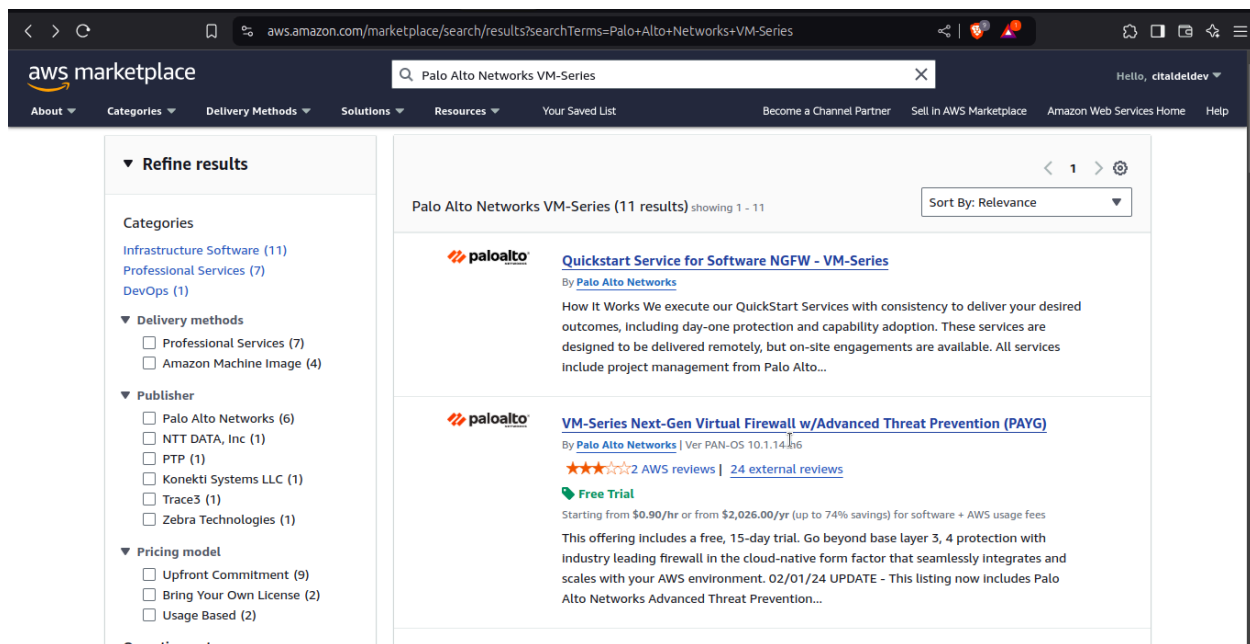
```
palo-alto-firewall/  
|  
├── main.tf  
├── variables.tf  
├── outputs.tf  
├── provider.tf  
├── network.tf  
├── firewall.tf  
├── security_groups.tf  
└── terraform.tfvars
```

Initial Steps

1. Login to your AWS account and create a User with programmatic access
 - Download the CSV file for the access keys and secrets
 - Ensure the User has required permissions to create AWS resources
2. EC2 Key Pair Setup
 - Go to EC2 instance page and create a new key pair
 - Save the key pair to your local machine
 - Grant key pair permissions:

```
chmod 400 your-key-pair.pem
```

3. AWS Marketplace Configuration
 - Search for "Palo Alto VM-Series Next-Gen Virtual Firewall w/Advanced Threat Prevention (PAYG)"



- Select Palo Alto VM-Series Next-Gen Virtual Firewall w/Advanced Threat Prevention (PAYG) and click on try for free. then go ahead to subscribe to get a 15-day free trial to use the firewall.

aws marketplace

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palooalto VM-Series Next-Gen Virtual Firewall w/Advanced Threat Prevention (PAYG) Info

Sold by: Palo Alto Networks

Free Trial

Protect AWS workloads, VDLs, and user traffic with AI/ML-powered virtual firewall from Palo Alto Networks. This listing includes VM-Series virtual firewall, Palo Alto Networks flagship - Advanced Threat Prevention security servi...

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Overview

VM-Series Virtual Firewalls:

Secure Your Business in Any Cloud

Highlights

- Best-in-class public cloud network security solution powered by AI/ML and latest threat research protects your workloads against day zero and known threats with application layer 7 visibility
- Integrations with Gateway Load Balancer, AWS Auto Scaling, and

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Learn more

Purchase order number - optional

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Software contract

Select contract option(s)

Total contract price \$0 Due now

Create Contract

- Select configuration:
 - Choose AMI type
 - Select Software Version
 - Choose AWS region

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Choose a fulfillment option and software version to launch this software.

Fulfillment option
64-bit (x86) Amazon Machine Image (AMI)

Software version
PAN-OS 10.1.14-h6 (Nov 20, 2024)

Region
US West (N. California)

Ami Id: ami-0a7d7460f53cb5b5b

Ami Alias: /aws/service/marketplace/prod-hhtxhwx3jg6k/pan-os-10.1.14-h6 [Learn More](#) **New**

Product Code: e9yfvj3uag5uo5j2hjkv74n

[Release notes \(updated November 20, 2024\)](#)

This is an estimate of typical software and infrastructure costs based on your configuration. Your actual charges for each statement period may differ from this estimate.

Software Pricing

VM-Series Next-Gen Virtual Firewall w/Advanced Threat Prevention (PAYG)	\$1.08 /hr OR \$4,049.00 /yr
---	------------------------------

Free Trial
running on c5n.xlarge

Infrastructure Pricing

EC2:	1 * c5n.xlarge
Monthly Estimate:	\$194.00/month

- Note down the AMI ID for Palo Alto firewall

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US West (N. California)

Ami Id: ami-0a7d7460f53cb5b5b

Ami Alias: /aws/service/marketplace/prod-hhtxhwx3jg6k/pan-os-10.1.14-h6 [Learn More](#) **New**

Product Code: e9yfvj3uag5uo5j2hjkv74n

[Release notes \(updated November 20, 2024\)](#)

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running on c5n.xlarge

Infrastructure Pricing

EC2:	1 * c5n.xlarge
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Configuration Steps

Step 1: AWS Credentials Setup

PS: you need to ensure you already have the AWS cli installed as well as Terraform in your local machine. if you don't, kindly go through this [AWS DOCS](#) to install CLI and this [TERRAFORM DOCS](#) to install terraform

```
# Configure AWS CLI
aws configure
# Enter your AWS Access Key ID
# Enter your AWS Secret Access Key
# Enter default region (e.g., us-west-2)
# Enter output format (json)

# Create S3 bucket for Terraform backend

aws s3api create-bucket --bucket palo-alto-unique123 --region us-west-1
--create-bucket-configuration LocationConstraint=us-west-1
```

Step 2: Terraform Provider Configuration (provider.tf)

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

backend "s3" {
  bucket = "palo-alto-unique123"
  key    = "palo-alto/terraform.tfstate"
  region = "us-west-1"
}

provider "aws" {
  region      = var.aws_region
  access_key  = var.aws_access_key
  secret_key  = var.aws_secret_key
}
```

Step 3: Network Configuration (network.tf)

Now lets configure the infrastructure network part bit by bit

- **VPC Creation** : This is the main network container for your AWS resources. It provides a virtual networking environment for your resources. inside the network.tf, add the following below

hcl

```
resource "aws_vpc" "firewall_vpc" {  
  cidr_block      = "10.0.0.0/16"  
  enable_dns_hostnames = true  
  tags = {  
    Name = "Palo Alto Firewall VPC"  
  }  
}
```

- **Internet Gateway** :add the following below inside the network.tf

```
resource "aws_internet_gateway" "firewall_igw" {  
  vpc_id = aws_vpc.firewall_vpc.id  
  tags = {  
    Name = "Firewall VPC Internet Gateway"  
  }  
}
```

- **Subnets (Management and Data)** :You can have multiple subnets in a VPC. Add the following below inside the network.tf

```
# Management Subnet (Private)
resource "aws_subnet" "management_subnet" {
  vpc_id            = aws_vpc.firewall_vpc.id
  cidr_block        = "10.0.1.0/24"
  availability_zone  = "us-west-1c"
  map_public_ip_on_launch = true
  tags = {
    Name = "Firewall Management Subnet"
  }
}

# Untrust/External Subnet (Public)
resource "aws_subnet" "untrust_subnet" {
  vpc_id            = aws_vpc.firewall_vpc.id
  cidr_block        = "10.0.2.0/24"
  availability_zone  = "us-west-1c"
  map_public_ip_on_launch = true
  tags = {
    Name = "Untrust Subnet"
  }
}

# Trust/Internal Subnet (Private)
resource "aws_subnet" "trust_subnet" {
  vpc_id            = aws_vpc.firewall_vpc.id
  cidr_block        = "10.0.3.0/24"
  availability_zone  = "us-west-1c"
  map_public_ip_on_launch = false
  tags = {
    Name = "Trust Subnet"
  }
}
```

- **Route Tables** : This is used to determine where traffic should be routed. Also add the following below inside the network.tf

```
# Management Route Table (Restricted)
resource "aws_route_table" "management_route_table" {
  vpc_id = aws_vpc.firewall_vpc.id
  route {
    cidr_block      = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.firewall_igw.id
  }
  tags = {
    Name = "Management Subnet Route Table"
  }
}

# Untrust Route Table (Public)
resource "aws_route_table" "untrust_route_table" {
  vpc_id = aws_vpc.firewall_vpc.id
  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.firewall_igw.id
  }
  tags = {
    Name = "Untrust Subnet Route Table"
  }
}

# Trust Route Table (Private)
resource "aws_route_table" "trust_route_table" {
  vpc_id = aws_vpc.firewall_vpc.id
  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.firewall_igw.id
  }
  tags = {
    Name = "Trust Subnet Route Table"
  }
}
```


- Associate the Route Tables with the Subnets - Add the following into the network.tf file also

```
# Route Table Associations
resource "aws_route_table_association" "management_subnet_association" {
  subnet_id      = aws_subnet.management_subnet.id
  route_table_id = aws_route_table.management_route_table.id
}

resource "aws_route_table_association" "untrust_subnet_association" {
  subnet_id      = aws_subnet.untrust_subnet.id
  route_table_id = aws_route_table.untrust_route_table.id
}

resource "aws_route_table_association" "trust_subnet_association" {
  subnet_id      = aws_subnet.trust_subnet.id
  route_table_id = aws_route_table.trust_route_table.id
}
```

- **Elastic Network Interfaces** : Create Elastic Network Interface for both Management and Data Plane - Add the following below inside the network.tf

```
# Network Interfaces for Palo Alto Firewall
resource "aws_network_interface" "management_interface" {
  subnet_id    = aws_subnet.management_subnet.id
  private_ips  = ["10.0.1.10"]
  security_groups = [aws_security_group.firewall_management_sg.id]
  tags = {
    Name = "Firewall Management Interface"
  }
}

resource "aws_network_interface" "untrust_interface" {
  subnet_id    = aws_subnet.untrust_subnet.id
  private_ips  = ["10.0.2.10"]
  security_groups = [aws_security_group.firewall_management_sg.id]
  tags = {
    Name = "Firewall Untrust Interface"
  }
}

resource "aws_network_interface" "trust_interface" {
  subnet_id    = aws_subnet.trust_subnet.id
  private_ips  = ["10.0.3.10"]
  security_groups = [aws_security_group.firewall_management_sg.id]
  tags = {
    Name = "Firewall Trust Interface"
  }
}
```

Step 4: Security Group Configuration

Create a security group for the management interface and another for the data interface. These security groups will control the inbound and outbound traffic for each interface

- Create a new file named **security_groups.tf** and add the following code

```
resource "aws_security_group" "firewall_management_sg" {
  name           = "Palo Alto Management SG"
  description    = "Security group for Palo Alto firewall management interface"
  vpc_id        = aws_vpc.firewall_vpc.id

  ingress {
    from_port     = 22
    to_port       = 22
    protocol       = "tcp"
    cidr_blocks   = ["0.0.0.0/0"]
  }

  ingress {
    from_port     = 443
    to_port       = 443
    protocol       = "tcp"
    cidr_blocks   = ["0.0.0.0/0"]
  }

  egress {
    from_port     = 0
    to_port       = 0
    protocol       = "-1"
    cidr_blocks   = ["0.0.0.0/0"]
  }

  tags = {
    Name = "Firewall Management Security Group"
  }
}
```

- basically, what we did is to create a security group that allows inbound traffic on port 22 (SSH) and port 443 and outbound Traffic on all ports

Step 5: Firewall Deployment Configuration

Deploy AWS instance for Palo Alto Firewall with:

- Specific AMI
- Instance Type
- Network Interfaces

Create a new file named **firewall.tf** and add the following code

```
resource "aws_instance" "palo_alto_firewall" {
  ami          = var.palo_alto_ami
  instance_type = var.firewall_instance_type
  key_name      = "devops"

  network_interface {
    network_interface_id = aws_network_interface.management_interface.id
    device_index          = 0
  }

  network_interface {
    network_interface_id = aws_network_interface.untrust_interface.id
    device_index          = 1
  }

  network_interface {
    network_interface_id = aws_network_interface.trust_interface.id
    device_index          = 2
  }

  tags = {
    Name = "Palo Alto Firewall"
  }
}
```

- ☐ This code creates an AWS instance for the Palo Alto firewall, using the palo_alto_ami which we earlier got from AWS marketplace
- ☐ We also specify the instance type and key name for the instance

- Also set up a Network Load balancer. This serves several purposes, particularly for ensuring high availability, fault tolerance, and efficient handling of traffic directed to your **Palo Alto Firewall** instance. Add the following Inside your **firewall.tf** file

```
# Network Load Balancer (NLB) Configuration
resource "aws_lb" "firewall_nlb" {
  name           = "firewall-nlb"
  load_balancer_type = "network"
  subnets       = [aws_subnet.untrust_subnet.id]

  tags = {
    Name = "Palo Alto Firewall NLB"
  }
}

# Network Load Balancer Target Group (For the Firewall's Management Interface)
resource "aws_lb_target_group" "firewall_nlb_tg" {
  name        = "firewall-nlb-tg"
  port        = 443
  protocol    = "TCP"
  target_type = "instance"
  vpc_id      = aws_vpc.firewall_vpc.id

  health_check {
    protocol          = "TCP"
    port              = 443
    interval          = 30
    timeout           = 10
    healthy_threshold = 3
    unhealthy_threshold = 3
  }
}
```

```

tags = {
  Name = "Firewall NLB Target Group"
}

# Network Load Balancer Listener for HTTPS Traffic
resource "aws_lb_listener" "firewall_nlb_listener" {
  load_balancer_arn = aws_lb.firewall_nlb.arn
  port              = 443    # Listener for HTTPS traffic
  protocol          = "TCP"

  default_action {
    type             = "forward"
    target_group_arn = aws_lb_target_group.firewall_nlb_tg.arn
  }
}

# Attach the Palo Alto Firewall instance to the NLB Target Group
resource "aws_lb_target_group_attachment" "firewall_nlb_attachment" {
  target_group_arn = aws_lb_target_group.firewall_nlb_tg.arn
  target_id        = aws_instance.palo_alto_firewall.id
  port             = 443    # Management port (HTTPS)
}

```

Step 6: Variables Configuration (variables.tf)

Define variables for:

- AWS Region
- Access Keys
- AMI ID
- Instance Type

Create a new file named variables.tf and add the following code

```

variable "aws_region" {

```

```

description = "AWS deployment region"
type        = string
default     = "us-west-1"
}

variable "aws_access_key" {
  description = "AWS Access Key"
  type        = string
  sensitive   = true
}

variable "aws_secret_key" {
  description = "AWS Secret Key"
  type        = string
  sensitive   = true
}

variable "palo_alto_ami" {
  description = "Palo Alto VM-Series AMI ID"
  type        = string
}

variable "firewall_instance_type" {
  description = "EC2 Instance type for Palo Alto Firewall"
  type        = string
  default     = "m5.xlarge"
}

```

- Finally, you can add these into your **terraform.tfvars**

```

palo_alto_ami = "Replace this with the particular ami you got for palo alto
firewall "
aws_access_key = "add your aws access key"
aws_secret_key = "add your aws secret"

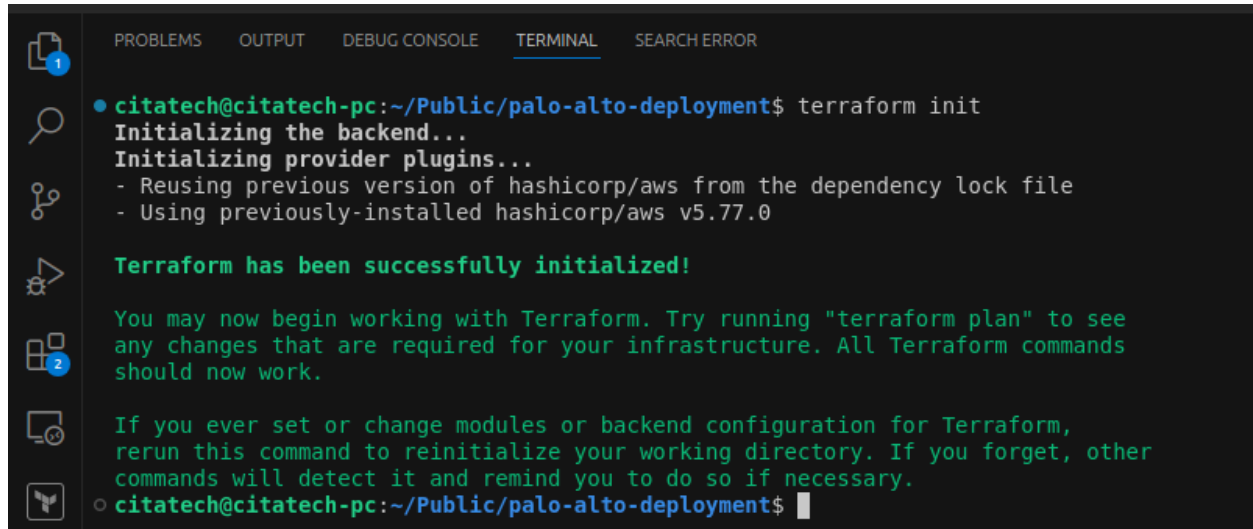
```

Step 7: 🚀 Deployment Commands

Now, lets deploy our infrastructure

- Initialize terraform

```
terraform init
```



```
citatech@citatech-pc:~/Public/palo-alto-deployment$ terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v5.77.0

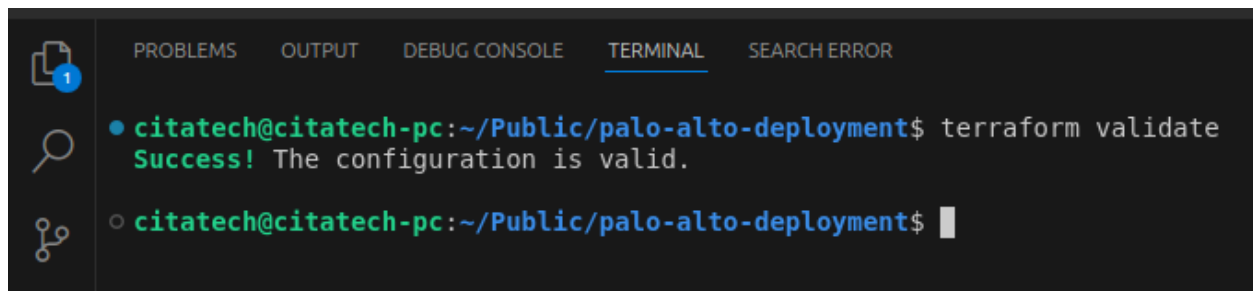
Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
citatech@citatech-pc:~/Public/palo-alto-deployment$
```

- Validate Configuration

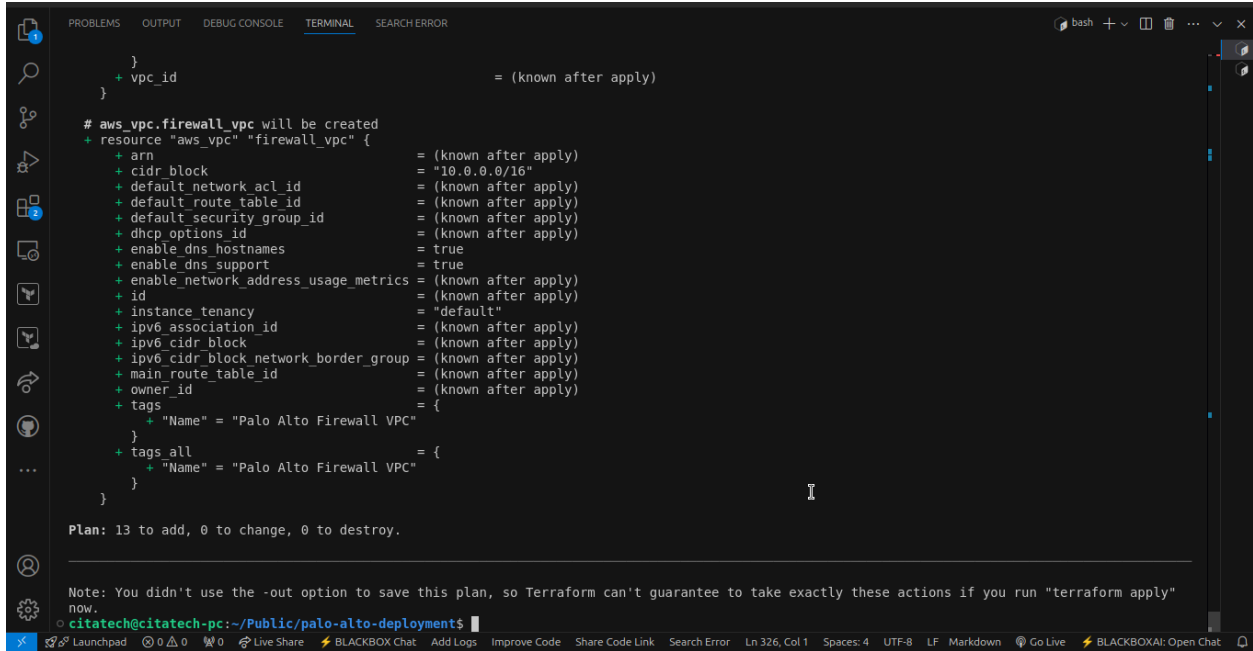
```
terraform validate
```



```
citatech@citatech-pc:~/Public/palo-alto-deployment$ terraform validate
Success! The configuration is valid.
citatech@citatech-pc:~/Public/palo-alto-deployment$
```

- Plan Deployment

```
terraform plan
```

```
    }
  + vpc_id                                = (known after apply)
}

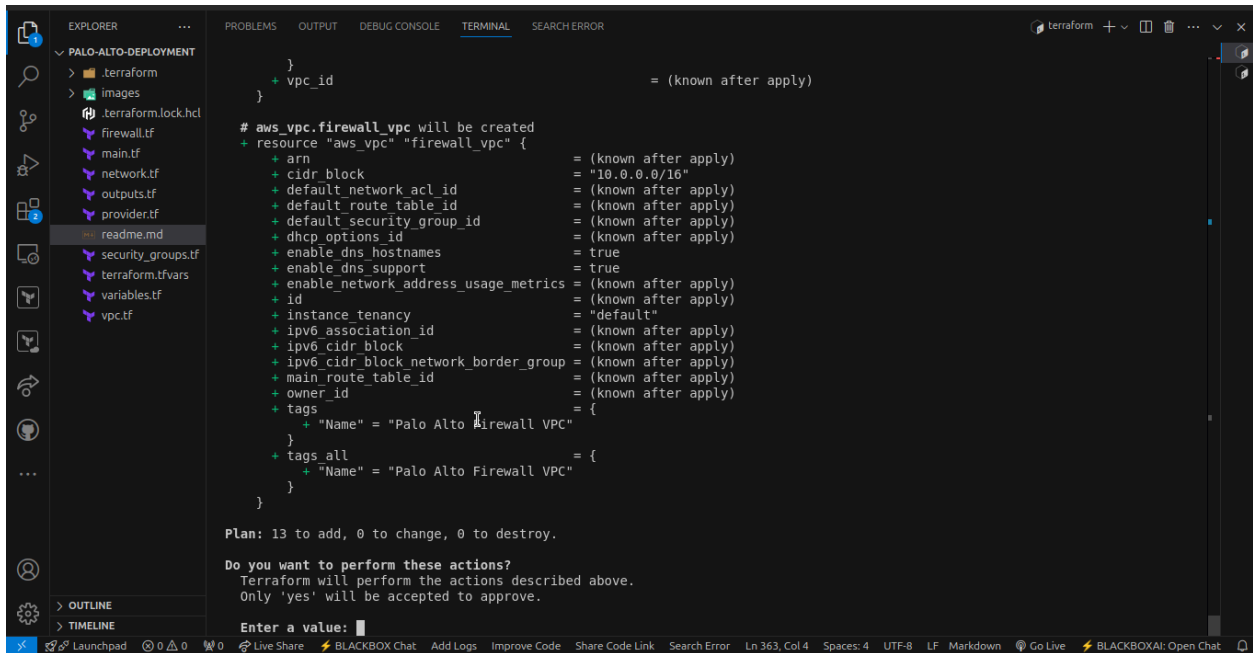
# aws_vpc.firewall_vpc will be created
+ resource "aws_vpc" "firewall_vpc" {
  + arn                                = (known after apply)
  + cidr_block                         = "10.0.0.0/16"
  + default_network_acl_id            = (known after apply)
  + default_route_table_id            = (known after apply)
  + default_security_group_id         = (known after apply)
  + dhcp_options_id                   = (known after apply)
  + enable_dns_hostnames               = true
  + enable_dns_support                 = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                                 = (known after apply)
  + instance_tenancy                   = "default"
  + ipv6_association_id                = (known after apply)
  + ipv6_cidr_block                    = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id                = (known after apply)
  + owner_id                           = (known after apply)
  + tags                               = {
    + "Name" = "Palo Alto Firewall VPC"
  }
  + tags_all                           = {
    + "Name" = "Palo Alto Firewall VPC"
  }
}

Plan: 13 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
citatech@citatech-pc:~/Public/palo-alto-deployments$
```

- Apply Configuration

terraform apply



```
    }
  + vpc_id                                = (known after apply)
}

# aws_vpc.firewall_vpc will be created
+ resource "aws_vpc" "firewall_vpc" {
  + arn                                = (known after apply)
  + cidr_block                         = "10.0.0.0/16"
  + default_network_acl_id            = (known after apply)
  + default_route_table_id            = (known after apply)
  + default_security_group_id         = (known after apply)
  + dhcp_options_id                   = (known after apply)
  + enable_dns_hostnames               = true
  + enable_dns_support                 = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                                 = (known after apply)
  + instance_tenancy                   = "default"
  + ipv6_association_id                = (known after apply)
  + ipv6_cidr_block                    = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id                = (known after apply)
  + owner_id                           = (known after apply)
  + tags                               = {
    + "Name" = "Palo Alto Firewall VPC"
  }
  + tags_all                           = {
    + "Name" = "Palo Alto Firewall VPC"
  }
}

Plan: 13 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: 
```

The terminal window displays the output of a Terraform run. It shows the creation of various AWS resources including VPCs, subnets, route tables, and security groups. The output is as follows:

```
Only 'yes' will be accepted to approve.
Enter a value: yes
aws_vpc.firewall_vpc: Creating...
aws_vpc.firewall_vpc: Still creating... [10s elapsed]
aws_vpc.firewall_vpc: Still creating... [20s elapsed]
aws_vpc.firewall_vpc: Creation complete after 21s [id=vpc-0f1cc8069d6daf481]
aws_route_table.data_route_table: Creating...
aws_subnet.data_subnet: Creating...
aws_subnet.management_subnet: Creating...
aws_internet_gateway.firewall_igw: Creating...
aws_security_group.firewall_management_sg: Creating...
aws_subnet.data_subnet: Creation complete after 2s [id=subnet-0439b4bad63bec517]
aws_network_interface.data_interface: Creating...
aws_route_table.data_route_table: Creation complete after 2s [id=rtb-0806a47525264f25a]
aws_route_table_association.data_subnet_association: Creating...
aws_internet_gateway.firewall_igw: Creation complete after 3s [id=igw-07f93be1666f693fb]
aws_route_table.management_route_table: Creating...
aws_route_table_association.data_subnet_association: Creation complete after 1s [id=rtbassoc-0a6f29796dfdf16d4]
aws_network_interface.data_interface: Creation complete after 2s [id=eni-0d700b29f924f923d]
aws_route_table.management_route_table: Creation complete after 2s [id=rtb-0fa68dd42361c7704]
aws_security_group.firewall_management_sg: Creation complete after 6s [id=sg-0909bbd58992e3b9b]
aws_subnet.management_subnet: Still creating... [10s elapsed]
aws_subnet.management_subnet: Creation complete after 13s [id=subnet-01bd4389c90817f93]
aws_route_table_association.management_subnet_association: Creating...
aws_network_interface.management_interface: Creating...
aws_route_table_association.management_subnet_association: Creation complete after 1s [id=rtbassoc-08693da967137212f]
aws_network_interface.management_interface: Creation complete after 2s [id=eni-0a0c6565850c465c4]
aws_instance.palo_alto_firewall: Creating...
aws_instance.palo_alto_firewall: Still creating... [10s elapsed]
aws_instance.palo_alto_firewall: Creation complete after 16s [id=i-08d41dfc3b4f1b5d0]
aws_eip.management_eip: Creating...
aws_eip.management_eip: Creation complete after 3s [id=eipalloc-023cd03ba3a29c99]

Apply complete! Resources: 13 added, 0 changed, 0 destroyed.
citatech@citatech-pc:~/Public/palo-alto-deployments$
```

- Confirm Infrastructure has been created

- VPC

The screenshot shows the AWS VPC dashboard for the us-west-1 region. It lists two VPCs: a default VPC and the Palo Alto Firewall VPC. The Palo Alto Firewall VPC is in an 'Available' state with a public IP address of 10.0.0.0/16.

Name	VPC ID	State	Block Public...	IPv4 CIDR
-	vpc-0e2fa06de1066d347	Available	Off	172.31.0.0/16
Palo Alto Firewall VPC	vpc-0d944e2cba6a0fd82	Available	Off	10.0.0.0/16

- Subnets

The screenshot shows the AWS VPC console with the 'Subnets (5)' page selected. The left sidebar contains the 'VPC dashboard' and 'Virtual private cloud' sections. The main area displays a table of subnets with columns for Name, Subnet ID, State, VPC, and Block Public IP. All subnets are in an 'Available' state.

Name	Subnet ID	State	VPC	Block Public IP
Untrust Subnet	subnet-0a5833c227248e3b5	Available	vpc-0508fcfe0ea864f9b Palo Alto Firewall	Off
-	subnet-01a22be4045c8dd0e	Available	vpc-0e2fa06de1066d347	Off
Trust Subnet	subnet-012ffc518795a89a5	Available	vpc-0508fcfe0ea864f9b Palo Alto Firewall	Off
-	subnet-0e40b85c057d742df	Available	vpc-0e2fa06de1066d347	Off
Firewall Management Subnet	subnet-0638011c90779e798	Available	vpc-0508fcfe0ea864f9b Palo Alto Firewall	Off

- Route Tables

The screenshot shows the AWS VPC console with the 'Route tables (5)' page selected. The left sidebar contains the 'VPC dashboard' and 'Virtual private cloud' sections. The main area displays a table of route tables with columns for Name, Route table ID, Explicit subnet associations, Edge associations, and Main. The 'Main' column indicates whether the route table is the primary route table for its VPC.

Name	Route table ID	Explicit subnet associations	Edge associations	Main
Management Subnet Route Table	rtb-057af6e1250a19f50	subnet-0638011c90779e798	-	No
Untrust Subnet Route Table	rtb-0a45752ace7b74a6a	subnet-0a5833c227248e3b5	-	No
-	rtb-0913cfec328f3fb83	2 subnets	-	Yes
Trust Subnet Route Table	rtb-0c49d5c5fbf406096	subnet-012ffc518795a89a5	-	No
-	rtb-0b35bb7ac01398741	-	-	Yes

- Internet Gateway

The screenshot shows the AWS VPC console with the 'Internet gateways' page selected. The left sidebar contains the 'VPC dashboard' and 'Virtual private cloud' sections. The main area displays a table of internet gateways with columns for Name, Internet gateway ID, State, and VPC ID. All gateways are in an 'Attached' state.

Name	Internet gateway ID	State	VPC ID
Firewall VPC Internet Gateway	igw-028d1e82b9414b1af	Attached	vpc-0d944e2cba6a0fd82 Palo Alto Firewall
-	igw-053f95e3ce3f5a287	Attached	vpc-0e2fa06de1066d347
Firewall VPC Internet Gateway	igw-07f93be1666f693fb	Attached	vpc-0f1cc8069d6daf481 Palo Alto Firewall

- Security Group

Find resources by attribute or tag				
☐	Name	Security group ID	Security group name	VPC ID
☐	-	sg-0aeaac362938a1f4a	default	vpc-0f1cc8069d6daf481
☐	-	sg-04804e13d6abddce6	default	vpc-0e2fa06de1066d347
☐	Firewall Manageme...	sg-909bbd58992e3b9b	Palo Alto Management SG	vpc-0f1cc8069d6daf481

- Palo Alto Firewall VM Instance

Instances (1) Info							
Last updated less than a minute ago							
Find Instance by attribute or tag (case-sensitive)							
All states							
running							
Clear filters							
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☐	Palo Alto Fire...	i-08d41dfc3b4f1b5d0	Running	m5.xlarge	Initializing	View alarms +	us-we

- Network Load Balancer

Load balancers (1)							
Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.							
Filter load balancers							
☐	Name	DNS name	State	VPC ID	Availability Zones	Type	
☐	firewall-nlb	firewall-nlb-0a0a68fde2b9...	Active	vpc-0508fcfe0ea864f9b	us-west-1c (usw1-az3)	network	



Post-Deployment Access

SSH into Firewall

- Create a **BASTION HOST** (ec2-instance), ssh into the instance by making use of an ssh agent

```
eval $(ssh-agent -s)

ssh-add /path/to/devops.pem

ssh -A ubuntu@<Bastion_Public_IP>
```

```
ubuntu@ip-10-0-1-86: ~ 135x32
cititech@cititech-pc:~/Downloads$ ssh -A ubuntu@54.193.125.27
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1016-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Nov 28 04:57:26 UTC 2024

System load:  0.0          Processes:      107
Usage of /:   23.0% of 6.71GB Users logged in:   1
Memory usage: 20%         IPv4 address for enX0: 10.0.1.86
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu Nov 28 04:31:02 2024 from 197.210.54.38
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-0-1-86:~$
```

- - Once inside the Bastion host, you can SSH into the firewall instance using your local SSH key:

```
ssh admin@<Firewall_Private_IP>
```

```
ubuntu@ip-10-0-1-86:~$ ssh admin@10.0.1.10
Last login: Wed Nov 27 20:31:55 2024 from 10.0.1.86

Number of failed attempts since last successful login: 1

There have been failed attempted logins from your username, which could mean someone is trying to brute-force your login. If this is not expected, consider contacting your system administrator.

admin@PA-VM> 
```

- Configure a new password, using the following command and follow the onscreen prompts

Configure

```
set mgt-config users admin password
```

- Set DNS server (if needed)

```
set deviceconfig system dns-setting servers primary <ip_address>
```

```
admin@PA-VM> configure
Entering configuration mode
[edit]
admin@PA-VM# set mgt-config users admin password
Enter password :
Confirm password :

[edit]
admin@PA-VM# set device
> device-object device-object
> deviceconfig deviceconfig

admin@PA-VM# set deviceconfig system dns-setting servers primary 52.52.253.110

[edit]
admin@PA-VM# 
```

- Commit changes

```
Commit
```

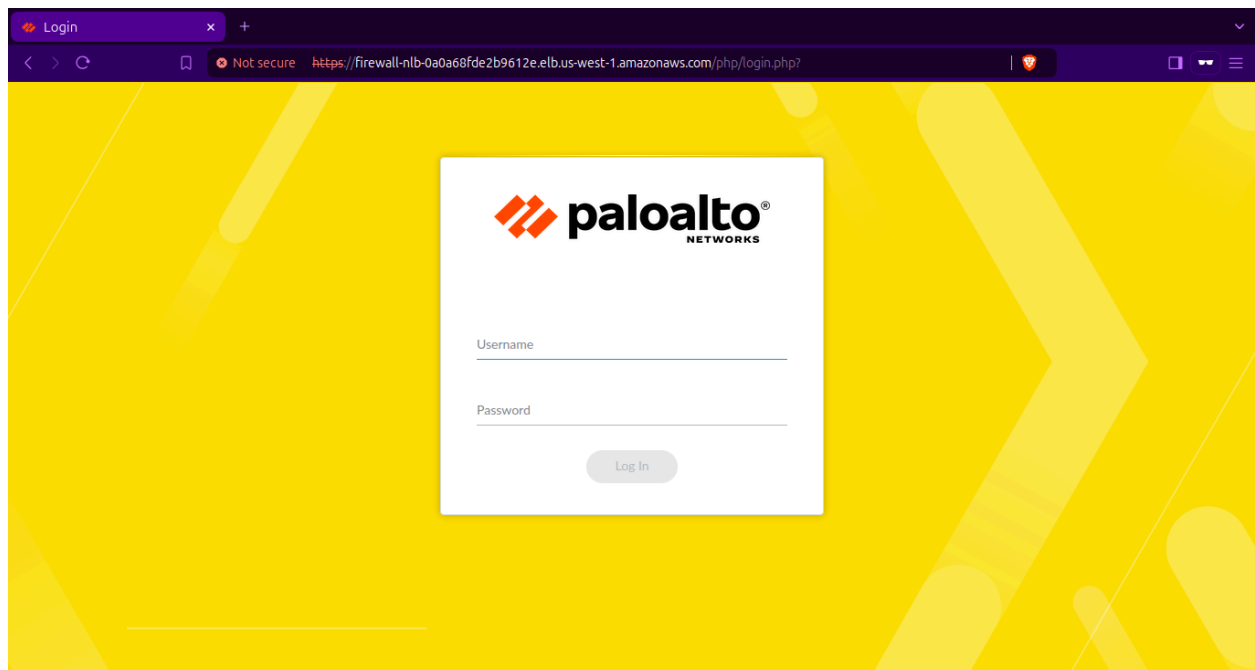
```
[edit]
admin@PA-VM# commit

Commit job 2 is in progress. Use Ctrl+C to return to command prompt
.....75%98%.....100%
Configuration committed successfully
Warning: Advance Routing mode is disabled , feature not supported
(Module: device)

[edit]
admin@PA-VM#
```

Finally, Access the web interface using

- URL: `https://<network-load-balancer-dns-name>`



- Username: admin
- Password: Your configured password
- Now you have gained access into the dashboard of your palo alto firewall

PA-VM - 61460

PA-VM

Not secure https://firewall-nlb-0a0a68fde2b9612e.elb.us-west-1.amazonaws.com/?#dashboard:vsys1

PA-VM DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK DEVICE

Layout 3 Columns Widgets Last updated 05:34:24 5 mins

General Information

Device Name: PA-VM

MGT IP Address: 10.0.1.10 (DHCP)

MGT Netmask: 255.255.255.0

MGT Default Gateway: 10.0.1.1

MGT IPv6 Address: unknown

MGT IPv6 Link Local Address: fe80::443:70ff:fe0fb34b/64

MGT IPv6 Default Gateway: 06:43:70:0fb3:4b

Model: PA-VM

Serial #: 44853D0AF155BD2

CPU ID: AWSMP-04718B26AB5AB10C5:e9yvyj3uag5uo5j2hjkv74ncus-west-1

UUID: EC222BA8-4EDF-EC41-FC33-4663D1AE5A27

VM Cores: 4

VM Memory: 15942152

VM License: VM-300

VM Capacity: 9.0 GB

VM Mode: Amazon AWS

Logged In Admins

Admin	From	Client	Session Start	Idle For
admin	10.0.1.86	CLI	11/27/2024 20:31:57	00:00:47s
admin	197.210.54.38	Web	11/27/2024 20:34:04	00:00:17s

Data Logs

No data available.

System Logs

Description	Time
User admin logged in via Web from 197.210.54.38 using https	11/27 20:34:04
authenticated for user 'admin'. From: 197.210.54.38.	11/27 20:34:04
Commit job succeeded. Completion time=2024/11/27 20:33:36. Jobid=2. User=admin	11/27 20:33:36
Port MGT: Up auto	11/27 20:33:25
DNS Proxy object: mgmt-obj inherited following values from dynamic interface: mgmt-if: Primary DNS: 10.0.0.2 Secondary DNS: ::	11/27 20:33:25
DHCP BOUND: interface eth0, ip 10.0.1.10 netmask 255.255.255.0 dhcp server: 10.0.1.1	11/27 20:33:25
DNS Proxy object: mgmt-obj inherited following values from dynamic interface: mgmt-if: Primary DNS: :: Secondary DNS: ::	11/27 20:33:24
Port MGT: Down auto	11/27 20:33:24

Config Logs

Command	Path	Admin	Time
commit		admin	11/27 20:33:01
set	config mgmt-config users	admin	11/27 20:32:58

Locks

No locks found

ACC Risk Factor (Last 60 minutes)

0.0

admin | Logout | Last Login Time: Not available | Session Expire Time: 12/27/2024 20:34:04

Tasks | Language | Alarms

paloalto

PA-VM

Not secure https://firewall-nlb-0a0a68fde2b9612e.elb.us-west-1.amazonaws.com/?#policies:vsys1:policies/security-rulebase

PA-VM DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK DEVICE

Security

NAT

QoS

Policy Based Forwarding

Decryption

Tunnel Inspection

Application Override

Authentication

DoS Protection

SD-WAN

Policy Optimizer

New App Viewer

Rules Without App Controls

Unused Apps

Rule Usage

Unused in 30 days

Unused in 90 days

Unused

NAME	TAGS	TYPE	Source				Destination				APPLICATION	SERV
			ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE			

Object: Addresses

+ Add Delete Clone Override Revert Enable Disable Move

PDF/CSV Highlight Unused Rules View Rulebase as Groups Reset Rule Hit Counter Group

https://firewall-nlb-0a0a68fde2b9612e.elb.us-west-1.amazonaws.com/?#

Tasks | Language

paloalto



Common Issues:

- Incorrect AMI ID
- Network Interface Configuration
- Security Group Restrictions
- Subnet CIDR Conflicts

Verification Commands:

```
# Verify Instance Status
aws ec2 describe-instances --filters "Name=tag:Name,Values=Palo Alto Firewall"

# Check Network Interfaces
aws ec2 describe-network-interfaces
```



Cleanup

```
# Destroy Infrastructure
terraform destroy
```